Absolute figure and table placement with the absfigure package

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Contents

NOTE This is version 0.2 of absfigure. The package is functional but should be considered experimental. Please report any issues.

1 Introduction

LaTeX's float mechanism (`\begin{figure}[t/b/h/p]`) provides only *relative* placement hints. There is no built-in way to say "put this figure at the bottom of page 6, column 2." The absfigure package provides exactly that capability for both figures and tables through a simple key-value interface.

Floats placed with absfigure integrate with LaTeX's page builder: text correctly reflows around them, `\caption`, `\label`, and `\ref` all work normally, and ordinary floats continue to function alongside absolute ones. Both `\listoffigures` and `\listoftables` pick up the placed floats automatically.

1.1 Requirements

- LaTeX 2_ε (1995/12/01 or later)
- The keyval package (part of the graphics bundle)
- The environ package (available on CTAN)

1.2 Loading the package

```
\usepackage{absfigure}
```

No package options are currently defined.

2 User interface

2.1 The `absfigure` environment

`\begin{absfigure}[`⟨*keys*⟩`]`
⟨*figure body*⟩
`\end{absfigure}`

Places a single-column figure on a specific page and column. The ⟨*figure body*⟩ may contain `\includegraphics`, `\caption`, `\label`, and any other content you would normally put inside a LaTeX `figure` environment.

EXAMPLE

```
\begin{absfigure}[page=6, pos=b, col=2]
  \includegraphics[width=\columnwidth]{fig.pdf}
  \caption{My figure}
  \label{fig:myfig}
\end{absfigure}
```

2.2 The `absfigure*` environment

`\begin{absfigure*}[`⟨*keys*⟩`]`
⟨*figure body*⟩
`\end{absfigure*}`

Places a spanning (double-column) figure, analogous to LaTeX's `figure*`. The `col` key is ignored for this environment since the figure spans both columns.

EXAMPLE

```
\begin{absfigure*}[page=3, pos=t]
  \includegraphics[width=\textwidth]{wide.pdf}
  \caption{A wide figure}
\end{absfigure*}
```

2.3 The abstable environment

`\begin{abstable}[`⟨*keys*⟩`]`
⟨*table body*⟩
`\end{abstable}`

Places a single-column table on a specific page and column. The ⟨*table body*⟩ may contain `tabular`, `\caption`, `\label`, and any other content you would normally put inside a LaTeX `table` environment. The same keys (`page`, `pos`, `col`) are accepted.

EXAMPLE

```
\begin{abstable}[page=4, pos=t, col=1]
  \centering
  \caption{My table}
  \label{tab:mytab}
  \begin{tabular}{lcc}
    \hline
    Item  & Value & Unit \\
    \hline
    Alpha & 1.0   & m/s  \\
    \hline
  \end{tabular}
\end{abstable}
```

2.4 The abstable* environment

`\begin{abstable*}[`⟨*keys*⟩`]`
⟨*table body*⟩
`\end{abstable*}`

Places a spanning (double-column) table, analogous to LaTeX's `table*`. The `col` key is ignored for this environment since the table spans both columns.

EXAMPLE

```
\begin{abstable*}[page=5, pos=b]
  \centering
  \caption{A wide table}
  \begin{tabular}{lccccc}
    \hline
    Exp & T1 & T2 & T3 & Mean & Std \\
    \hline
    A   & 4.2 & 4.5 & 4.1 & 4.27 & 0.21 \\
    \hline
  \end{tabular}
\end{abstable*}
```

3 Keys

3.1 page

page=⟨*integer*⟩ — Target page number (positive integer).
page=current — Place on the page currently being built (default).
page=end — Place on an extra page appended after the document.

If page is omitted, it defaults to current — the float is placed on the page where the environment appears in the source.

An error is raised if page is zero, negative, or non-numeric (other than the special values current and end).

If the target page number exceeds the number of pages in the document, the float is never placed and a warning is raised at the end of the LaTeX run.

WARNING If a float targets a page that has already been shipped out in the current run, it cannot be placed until the next run. In that case, rerun LaTeX after editing. The package writes a placement cache and emits a warning when absolute-float placement is not yet final.

3.2 pos

pos=t — Place at the top of the page (default).
pos=b — Place at the bottom of the page.

The aliases pos=top and pos=bottom are also accepted.

An error is raised if pos is set to anything other than t/top or b/bottom.

NOTE The LaTeX kernel does not natively support bottom-placed spanning floats (there is no `\@dblbotlist`). The absfigure package implements its own spanning-bottom mechanism, so pos=b works with all four environments (`absfigure`, `absfigure*`, `abstable`, `abstable*`).

3.3 col

col=1 — Place in the first (left) column (default).
col=2 — Place in the second (right) column.

The aliases col=l/col=left and col=r/col=right are also accepted.

This key is only meaningful in `\twocolumn` mode. In single-column documents, col=2 (or col=right) raises an error.

For starred environments (`absfigure*`, `abstable*`), this key is silently ignored (the float always spans both columns).

An error is raised if col is set to anything other than 1/left or 2/right.

4 Captions, labels, and cross-references

`\caption` and `\label` work normally inside all four environments. They are executed at *placement* time (during the output routine), so:

- The figure/table counter is incremented in the order floats are actually placed, not the order they appear in the source.
- `\ref` and `\pageref` resolve correctly after two compilations (standard LaTeX behaviour for cross-references).
- Figures appear in `\listoffigures` and tables in `\listoftables` automatically.

5 Error handling

The package performs validation on all keys and issues errors or warnings as appropriate:

`page` **omitted** Defaults to `current` (the page being built). No error.

`page` **is zero, negative, or non-numeric** **Error:** `Invalid page value '⟨value⟩'`. The `page` key requires a positive integer, `current`, or `end`.

`page` **exceeds total pages** **Warning** at end of run: `Float N (figure/table, target page=X) was not placed in this run`. If the warning persists after rerunning LaTeX, the target page or slot is unreachable.

`page=current` Valid. Resolves to the page number being built at definition time.

`page=end` Valid. An extra page is appended after the document's last page, and the float is placed there.

float targets an earlier page than its position in the source Valid. Rerun LaTeX so the cached float definition can be replayed before that page is built.

`pos` **is not** `t`/`top` **or** `b`/`bottom` **Error:** `Invalid pos value '⟨value⟩'`.

`col` **is not** `1`/`l`/`left` **or** `2`/`r`/`right` **Error:** `Invalid col value '⟨value⟩'`.

`col=2` **or** `col=right` **in single-column mode** **Error:** `col=2 (right) used in single-column mode`.

6 How it works

Understanding the internal mechanism may be helpful for advanced users and for debugging.

6.1 Core mechanism

LaTeX calls `\@startcolumn` at the start of every column (and page). The package patches this command to inject absolute figures *before* deferred floats are processed:

```
\@startcolumn (patched):
  1. \global\@colroom\@colht      % reset height
  2. detect new column vs re-entry
  3. \absf@inject@figures          % OUR HOOK
  4. \@colroom -= \absf@colreduce  % apply reduction
  5. \@tryfcolumn / deferred       % original logic
```

6.2 Re-entry handling

The kernel may call `\@startcolumn` multiple times for the same column (via `\@whilesw\if@fcolmade`). Each re-entry resets `\@colroom`, undoing any reductions. The package tracks the current column identity (page number + column flag) and stores the cumulative space reduction in a dimension register (`\absf@colreduce`). On re-entry, injection is skipped but the stored reduction is re-applied.

6.3 Injection logic

For each pending float, the package checks whether the current page counter (`\c@page`) matches the target page and whether `\if@firstcolumn` matches the target column. On a match:

1. A float box register is allocated from `\@freelist`.
2. The captured body tokens are typeset inside it (with `\@captype` set to `figure` or `table` so `\caption`/`\label` work and counters increment correctly).
3. `\count\@currbox` is set to encode the float type and placement using the kernel's `\ftype@`⟨*type*⟩ multiplier.
4. The box is added to `\@toplist` or `\@botlist` and the space reduction is accumulated in `\absf@colreduce`.

For spanning floats (`absfigure*`, `abstable*`), the box is added to `\@dbltoplist` and `\@colht` is reduced instead (this reduction naturally survives re-entries since `\@startcolumn` resets `\@colroom` from `\@colht`).

If multiple absolute floats target the same slot (`page`, `col`, and `pos`), they are placed in source order. A later float does not overtake an earlier one in the same slot.

6.4 Why text reflows correctly

Reducing `\@colroom` inside `\@startcolumn` propagates to `\vsize`, which the TeX page builder uses to determine where to break the column. Less available height means less text on the page, leaving room for the figure.

7 Limitations and known issues

- If a float targets a page that is already past in the current run, an additional LaTeX run is needed to finalize placement. This is expected and works similarly to other multi-pass features such as cross-references and bibliographies.
- If many absolute floats target the same page, the available text height may become negative and TeX will produce an overfull vbox. Plan float sizes accordingly.
- Float numbering follows placement order, not source order. If you define figure A (targeting page 5) before figure B (targeting page 2), figure B will get a lower number. The same applies to tables.
- Cross-references (`\ref`, `\pageref`) require two compilations to resolve, as is standard for LaTeX.
- The package has not been tested with the multicol package. It is designed for LaTeX's built-in `\twocolumn` mechanism.

8 Complete example

```
\documentclass[twocolumn]{article}
\usepackage{graphicx}
\usepackage{lipsum}
\usepackage{absfigure}

\begin{document}
```

```
\begin{absfigure}[page=2, pos=t, col=1]
  \centering
  \includegraphics[width=\columnwidth]{fig1.pdf}
  \caption{Top of page 2, column 1}
  \label{fig:one}
\end{absfigure}

\begin{absfigure*}[page=3, pos=t]
  \centering
  \includegraphics[width=\textwidth]{wide.pdf}
  \caption{Spanning figure on page 3}
  \label{fig:wide}
\end{absfigure*}

% A table on page 2, bottom of column 2
\begin{abstable}[page=2, pos=b, col=2]
  \centering
  \caption{Results}
  \label{tab:results}
  \begin{tabular}{lcc}
    \hline
    Method & Accuracy & F1 \\
    \hline
    Ours   & 94.2     & 0.91 \\
    \hline
  \end{tabular}
\end{abstable}

\begin{absfigure}[page=end]
  \centering
  \includegraphics[width=\columnwidth]{extra.pdf}
  \caption{Appended at the end}
  \label{fig:extra}
\end{absfigure}

See Figure~\ref{fig:one}, Table~\ref{tab:results},
and Figure~\ref{fig:extra}.

\lipsum[1-30]

\end{document}
```

9 Version history

v0.1 (2026/02/25) Initial release.

- `absfigure`, `absfigure*`, `abstable`, and `abstable*` environments
- Keys: `page`, `pos`, `col`
- `page=current` (default when `page` omitted)
- `page=end` support
- Validation of all keys with meaningful error messages
- End-of-document warnings for unplaced figures

v0.2 (2026-04-03) Documentation and placement update.

- Environments documented with optional-argument syntax
- Earlier-page placement works across reruns via a cache file
- End-of-run warning now tells the user to rerun LaTeX
- Floats targeting the same slot are placed in source order